

Ryde Technologies: Enhancing the customer experience and driving growth in new markets

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About Ryde Technologies

Founded as an on-demand carpooling app, Ryde Technologies provides a range of transportation services, including carpool, private hire and taxis, through about 60,000 drivers. Established in 2015, the business is restarting expansion plans into markets such as Malaysia it had paused during the pandemic.

Industries: Automotive

Google Cloud results

- Achieves availability of at least 99.9%, enabling the business to attract new customers and drivers
- Enables seamless migration of 1TB of data to multiple database instances, avoiding reputation-damaging disruption
- Supports chat and tracking services that enable customers to view driver locations and check on pickup times
- Enables the business to

Reduced
mess
respo
by 70

Location: Singapore

Cloud Storage

searce^o

About Searce

Searce is a niche cloud consulting business with futuristic tech in its DNA, focused on "realizing the Next in the Now" for its clients. Specializing in cloud data engineering, AI/ML, and advanced cloud InfraTech, such as Anthos and Kubernetes, Searce is one of the top partners for Google Cloud globally, with 3,000+ clients successfully moved to the cloud.

Google Cloud (<https://cloud.google.com/>)

BigQuery (<https://cloud.google.com/bigquery/>)

Cloud CDN (<https://cloud.google.com/cdn/>)

Cloud Spanner (<https://cloud.google.com/spanner/>)

Cloud SQL (<https://cloud.google.com/sql/>)

Cloud Storage (<https://cloud.google.com/storage/>)

Compute Engine

(<https://cloud.google.com/compute/>)

Cloud Firestore

(<https://cloud.google.com/firestore/>)

Dataflow (<https://cloud.google.com/dataflow/>)

determine the
prices
acceptable to
customers
and drivers,
and attract
new
customers

- Delivers
centralized
authentication
and single
sign-on
across Google
and non-
Google Cloud
services used
by the
business

Kubernetes Engine

(<https://cloud.google.com/kubernetes-engine/>)

Google Workspace

(<https://workspace.google.com/>)

Founded in 2015 as an on-demand carpooling app, Ryde Technologies (<https://www.rydesharing.com>) (Ryde) has grown to a portfolio of nine services, including private hire, delivery and pet transport. "We bring communities together and create sustainable value for our users by providing simple, seamless, and reliable transportation and logistics services at reasonable prices," explains Nitin Dolli, Chief Technology Officer, Ryde.

The business has more than 40 employees in Singapore and about 60,000 drivers divided between three categories, namely carpool, private hire and taxi drivers. Carpool drivers are legally restricted to

two trips per day while private hire and taxi drivers can take unlimited trips, but must take out commercial insurance and complete relevant courses and certifications.

Ryde initially launched its carpooling service in Hong Kong and Singapore based on a multi-cloud environment. However, when a global ride-hailing service departed the Singapore market in May 2018, the business adapted its expansion strategy to focus on Singapore, with soft launches into Malaysia and Australia.

The company also explored opportunities to add features such as live tracking and chat to fulfil the expectations of a digitally-savvy customer base accustomed to the experience provided by other online transportation and ecommerce services.

"We wanted our users to be able to view drivers' journeys towards them and estimated times of arrival, and enable them to ask questions online in circumstances such as delayed arrivals," says Dolli.

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Accelerating growth in the cloud

To execute its expansion plans, the business targeted services and platforms that could support its transition from a monolithic architecture to microservices that would enable faster updates and improve overall business agility.

Ryde explored the opportunities Kubernetes and Google Cloud (<https://cloud.google.com>) presented to scale applications seamlessly, update them quickly and efficiently, support its soft-launch program and deliver driver tracking and chat services.

In the end, multiple factors prompted Ryde to decide on Google Cloud as their core provider. "We wanted to consume an existing service and build our own applications on top of it. We also needed a database

that could support real-time communication rather than requiring us to develop a system ourselves," says Dolli.

"We also wanted to focus on the business and strategic aspects of expanding into new markets rather than the technical details. Because Google Cloud uses Kubernetes in its own infrastructure, we knew it could scale and sustain the loads our business would demand of it, and this helped us finalize our decision."

Furthermore, the Google Cloud user interface was very developer-friendly and its billing systems provided the granular insights the business needed to optimize its use of compute and other resources. This would help Ryde align costs with growth by consuming only the resources needed at any given time.

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Doubling speed of execution with Searce

The business decided to move from its multi-cloud architecture to a Google Cloud environment in Singapore and worked with Searce (<https://www.searce.com/>) on the migration. "It was seamless dealing with Google Cloud, but when Searce came into the picture, its team members doubled the speed of execution and increased the

pace of development," says Dolli. "I learned a lot working with both teams."

In particular, Ryde worked with Searce to determine that Firestore

(<https://firebase.google.com/docs/firestore>), a cloud-hosted NoSQL database, would support the location-based tracking and other features needed to deliver its tracking and chat services.

Ryde also liaised with Google Cloud teams to learn about Kubernetes, in particular how the container orchestration system could scale to support its regional expansion plans. "That really put us into a positive position to implement our new technology architecture," says Dolli. "With Google Kubernetes Engine (<https://cloud.google.com/kubernetes-engine>), we can focus on executing our business plans rather than monitoring our systems 24 hours a day, seven days a week, and relax knowing service uptime exceeds 99.9%."

Improving response speeds by 70% with Cloud CDN

The new architecture also incorporates a content delivery network to support its soft-launches into new markets. The business opted to deploy Cloud CDN (<https://cloud.google.com/cdn>) to access Google Cloud's worldwide network and address delays in communications to and from drivers in Australia in particular. "With Cloud CDN, we reduced response speeds by 70%, so when we soft-launched into Australia prior to the pandemic, drivers there had the same experience as drivers in Singapore," says Dolli.

Ryde now runs its SQL database across multiple Cloud SQL (<https://cloud.google.com/sql>) instances having elected to move it from Compute Engine (<https://cloud.google.com/compute>), an exercise that entailed shifting a terabyte of data without downtime or dropping a single request or query.

"Any issues with the transition would have compromised the user experience and damaged our image," says Dolli. "However, the ease of use, stability and technical expertise embedded in Google Cloud products enabled us to complete the migration without disruption." The business now holds driver data and documentation in a stable, resilient database environment that supports features and business activities such as user reviews, profiles, subscription systems, payments and bonuses.

Ryde is now working with Searce and Google Cloud to explore the potential of Cloud Spanner (<https://cloud.google.com/spanner>) to support rapid scalability and potential entry to new markets such as Europe, the United States or South America.



Exploring new business opportunities with BigQuery Analytics

Insights derived from analysis of data are key to Ryde's identification of new business opportunities and ways of improving its customer experience. The business is streaming location and user behavior data through Dataflow

(<https://cloud.google.com/dataflow>) to a BigQuery

(<https://cloud.google.com/bigquery>) data warehouse for analysis with an internal tool to achieve these objectives, as well as to repositories such as Cloud Storage (<https://cloud.google.com/storage>).

"The idea of using any analytical tool or segregating or channelling data is to decide what to do next," says Dolli. "We look at new features to develop, how to enhance the experience and where and why users may not be proceeding to book a trip. This is so we can review all the data points from a BigQuery dashboard and determine how we can attract more users than our competitors."

Pricing is a particularly key focus for the business as it largely determines whether a potential customer proceeds to book a ride and whether a driver accepts the trip. "There is a thin line between these two user segments and we use the BigQuery dashboards to make our pricing decisions," says Dolli. The business has also used insights enabled by BigQuery to execute a strategy that includes launching private hire and other services that target the premium market in Singapore.

"With BigQuery and Google Cloud, we can look at real-time supply and demand and how people react to time-sensitive marketing campaigns," says Dolli.

"We can also look at all other factors affecting decision-making. Ultimately we want to keep the ecosystem moving and keep users and drivers happy."

Navigating pandemic challenges with agility

The business completed proofs of concept for the new Google Cloud architecture in February 2019 and finalized the migration in September that year, giving the business the ability to adapt rapidly to changing market circumstances. When the pandemic struck, the business was able to pause its soft launches in Malaysia and Australia, while supporting demand in its home market that remained resilient despite government measures to minimize spread of the coronavirus.

"Our assumption was that demand would drop over this period, but we noticed the opposite happening,"

says Dolli. "When people could travel, they did not want to use public transport with others, so demand for private hire vehicles went through the roof. We also saw a lot of healthcare and other frontline workers taking trips with us. With Google Cloud, we could accommodate unexpected demand with no deterioration in performance."

With demand starting to stabilize at pre-pandemic levels, Ryde plans to restart in 2023 its paused launch into Malaysia, continue to build out its suite of services and focus on delivering a high quality experience for users and drivers.

The business complements its use of Google Cloud infrastructure and data services with a Google Workspace (https://workspace.google.com/intl/en_au/) deployment spanning all 40 employees. The workplace productivity and collaboration suite supports Ryde's growth plans through seamless integration with Google Cloud and the delivery of single sign-on that extends login and authentication beyond Google Workspace to third party applications.

"With Google Cloud, we're realizing our ambitions to grow our range of services and position ourselves to expand into the United States, Europe and Asia-Pacific markets with tightly managed investment in technology resources and a focus on business planning rather than infrastructure administration," says Dolli.

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